

Day 2 Write-Up: Preprocessing Choices and Model Comparison

Day 2 work focused on building a more robust classification pipeline for predicting whether an individual earns more than \$50K. The business goal is not simply to maximize overall accuracy, but to improve precision so that outreach is directed toward people who are most likely to be high earners. In that setting, false positives are more costly than false negatives, so precision is the primary metric and ROC AUC is used as a secondary, threshold-independent check on overall ranking quality.

Preprocessing choices

The pipeline prioritizes simplicity and interpretability for mixed data. We removed the redundant 'education' column and kept target labels separate. Numeric features (capital-gain, capital-loss, hours-per-week) used median imputation to handle skewness and outliers, followed by standard scaling to normalize inputs for models like logistic regression. Categorical features used most-frequent imputation and one-hot encoding to avoid false ordinal relationships, with `handle_unknown='ignore'` to safely manage unseen categories. This approach ensures robustness and compatibility across both linear and tree-based models.

Model comparison

We compared a majority-class and rule-based baseline against two machine learning models. The baselines performed poorly or provided only modest utility. Both ML models significantly outperformed these baselines, with Logistic Regression (Accuracy: 0.8528) and the Decision Tree (Accuracy: 0.8538) showing strong results.

Model	Accuracy	Precision	Recall	F1	ROC AUC	PR AUC
Majority-class	0.7607	0.0000	0.0000	0.0000	0.5000	0.2393
Rule-based	0.7520	0.4852	0.5967	0.5352	0.7493	0.5079
Logistic Regression	0.8528	0.7399	0.5937	0.6588	0.9032	0.7607

Model	Accuracy	Precision	Recall	F1	ROC AUC	PR AUC
Decision Tree	0.8538	0.7658	0.5607	0.6474	0.8823	0.6981

The Decision Tree achieves higher precision at the default threshold, making it attractive for conservative outreach. However, Logistic Regression excels in threshold-independent metrics, including ROC AUC (0.9032), indicating stronger overall ranking capability. While the tree offers slightly better precision at the operating point, the logistic model is more robust as a general ranking tool.

Recommendation

For this project, I would continue to evaluate both models, with Logistic Regression as the more interpretable and production-friendly baseline and the Decision Tree as a useful alternative that captures nonlinear relationships. The logistic model offers a strong balance of interpretability, ranking quality, and overall performance, while still delivering precision well above the rule-based baseline. Its coefficients also make it easier to explain to stakeholders and connect back to the earlier error analysis.

The Decision Tree remains valuable because it achieves slightly higher precision at the default threshold and may be attractive if we want a more conservative outreach policy. For that reason, it is reasonable to continue with both models and compare them further using threshold tuning, business cost assumptions, and stakeholder preferences.

In short, the preprocessing choices were appropriate for the problem and helped both models significantly outperform the simple baselines. If the goal is to move forward with a production-ready solution, both models are worth continuing to explore, with Logistic Regression as the leading candidate and the Decision Tree as a strong secondary option.